Case Study Report Group A.1

Research question: Does Annual Household Income Play a Role in Minority's Political Party Identification?

The reason for this research question is to understand whether annual household income is associated with political party identification among five ethnic minority groups, which are Indian, Pakistani, Bangladeshi, Black Caribbean, or Black African. Income is a major factor of socioeconomic position, which influences political preferences and political views. Minor ethnicities often experience different economic opportunities and political experiences compared to the major ethnicities, and examining variation within these groups helps avoid treating minorities as a homogeneous group. This research question might be of interest to:

- **Policy makers:** to understand how income level affects marginalized groups' political views and to create and design redistribution policies adjusted to their needs.
- **Social science researchers, political analysts:** translating current events into understandable narratives to interpret social behaviour, changes, and debate about polarization, economic inequality, or political behaviour and alignment
- **Political parties and campaign strategists:** enable them to benefit from the results to better analyse the election results, create future political strategies to attract voters, and make the right political strategies for the right target.
- **The general public:** it provides the public with a clearer understanding of what influences political choices and gives them a deeper insight into political and social issues.
- **Lobbies:** enable them to select effective arguments and resources toward efforts that are likely to yield tangible results.

From the "**UKDA-6970-stata**" folder, several files are available. To answer this research question, the most relevant files are:

- **BSE_2010.dta**, which contains the survey data collected for the 2010 UK election.
- **british_election_study_ethnic_minority_survey_2010_data_anon_ukda_data_dictionary**, which provides dataset's support documentation, including variable descriptions, value labels, and metadata necessary for interpreting the survey data.

Before conducting the analysis, several data preparation and transformation steps were performed to ensure the dataset was suitable for an analysis to answer the research question. The original dataset, BSE_2010.dta, was imported into RStudio, and the data dictionary was used to understand variable definitions and value labels.

There are many variables in the dataset and most of them are irrelevant to the research question; therefore, only relevant variables will be selected. These included Unique ID (Unqid), Ethnicity (zq101), Annual Household Income (zqinc), and Party Identification (bq9_1). These variables were extracted and stored in a new analysis dataset.

The dataset was then restricted to ethnic minority respondents by filtering the ethnicity variable to include only respondents identifying as Indian, Pakistani, Bangladeshi, Black

Caribbean, or Black African. Responses marked as Refused, Don't Know, or Not Stated for ethnicity, annual household income, and party identification were treated as missing and excluded from the dataset. Value labels were removed for analytical purposes while retaining the underlying numeric codes.

Lists of interested variables:

Filter variable:

zq101, labeled as "Ethnicity"
Nominal Categorical
Category lists:

Value = 7	Label = Asian or Asian British – Indian
Value = 8	Label = Asian or Asian British – Pakistani
Value = 9	Label = Asian or Asian British – Bangladeshi
Value = 11	Label = Black or Black British – Caribbean
Value = 12	Label = Black or Black British - African

Independent Variable (IV):

zqinc, labeled as "Annual Household Income"
Ordinal categorical, although this variable was labelled as nominal in the data dictionary, the annual household income are ordered from lowest to highest and are therefore treated as ordinal.
Category lists:

Value = 1	Label = 0 – 5000
Value = 2	Label = 5001 – 10000
Value = 3	Label = 10001- 15000
Value = 4	Label = 15001 – 20000
Value = 5	Label = 20001 – 25000
Value = 6	Label = 25001 – 30000
Value = 7	Label = 30001 – 35000
Value = 8	Label = 35001 – 40000
Value = 9	Label = 41000 – 45000
Value = 10	Label = 45001 – 50000
Value = 11	Label = 50001 – 60000
Value = 12	Label = 60001 – 70000
Value = 13	Label = 70001 – 80000
Value = 14	Label = 80001 or more

Dependent Variable (DV):

bq9_1, labeled as "Party Identification"
Nominal categorical
Category lists:

Value = 1	Label = None/No
Value = 2	Label = Labour

Value = 3	Label = Conservatives
Value = 4	Label = Liberal Democrats
Value = 5	Label = Scottish National Party (SNP)
Value = 6	Label = Plaid Cymru
Value = 7	Label = Green Party
Value = 8	Label = United Kingdom Independence Party (UKIP)
Value = 9	Label = British National Party (BNP)
Value = 10	Label = Coalition party/Conservative-Lib dem party
Value = 11	Label = Other
Value = 12	Label = Respect

Chi-square test was used to see whether there is an association between annual household income and political party identification. This is an appropriate analysis because Chi-square test to test two categorical variables, and both the independent variable, annual household income, and the dependent variable, party identification, are categorical variables. For this research question, it tests whether different annual household income groups play a role in political party identification among five ethnic minority groups. Importantly, this analysis tests for association rather than causation, which aligns with the exploratory nature of the research question.

To ensure that the Chi-square test results are reliable, three assumptions must be satisfied.

Lists of Assumptions:

1. **All Observations are independent**
2. **Both variables are categorical**
3. **At least 80% of Expected value of cells should be 5 or greater**

Two hypotheses are formulated to test the research question.

Lists of hypotheses:

Null hypothesis (H_0): There is no association between annual household income and political party identification among five ethnic minority groups.
Alternative hypothesis (H_1): There is an association between annual household income and political party identification among five ethnic minority groups.

After the chi-square test was performed, the result, shown in Figure 1, shows that the p-value is approximately 0.86, which is greater than 0.05. Therefore, the null hypothesis was not rejected, indicating that there is no association between annual household income and political party identification among five ethnic minority groups.

```
        Pearson's Chi-squared test

data:  table_income_party
X-squared = 100.86, df = 117, p-value = 0.8563
```

Figure 1: Chi-square Test Result 1

However, before relying on the results of the Chi-square test, it is necessary to check whether the test assumptions are satisfied, as can be see below.

1. **All Observations are independent:** In the BSE 2010 dataset, each respondent provides one observation, and these observations are independent of one another.
2. **Both variables are categorical:** Both the independent variable, annual household income, and the dependent variable, party identification, are categorical.
3. **At least 80% of Expected value of cells should be 5 or greater:** In this analysis, only around 30.71% of the expected counts are greater than five, meaning that the assumption is violated. Low expected counts can make the chi-square approximation less reliable and may affect both Type I (false positive) and Type II (false negative) error rates.

After checking the assumptions, it was found that the third assumption was violated; therefore, the result of the Chi-square test was not reliable. This occurred because the contingency table was overly large, containing many annual household income bands and political party categories. As a result, numerous sparse cells were created, with many cells containing very small counts.

To address this issue, both the independent variable (IV) and the dependent variable (DV) were adjusted. First, the annual household income bands were grouped into three broader categories: Low, Medium, and High. Second, political parties with very few responses were combined into a single category labelled “Others.”

After adjusting both variables, the third assumption was tested again and was now satisfied, with approximately 93.3% of the expected counts greater than five.

Results:

The Chi-square test was performed again. The new result, shown in Figure 2, shows that the p-value is approximately 0.000089, which is lesser than 0.05. Therefore, the null hypothesis was rejected, indicating that there is an association between annual household income and political party identification among five ethnic minority groups.

```
Pearson's Chi-squared test

data:  table_income_party
X-squared = 32.11, df = 8, p-value = 8.901e-05
```

Figure 2: Chi-square Test Result 2

Since the Chi-square test indicates that there is an association between annual household income and political party identification, Cramér's V was calculated to measure the strength of this association. The result of Cramér's V is 0.1031 indicates that the association is weak.

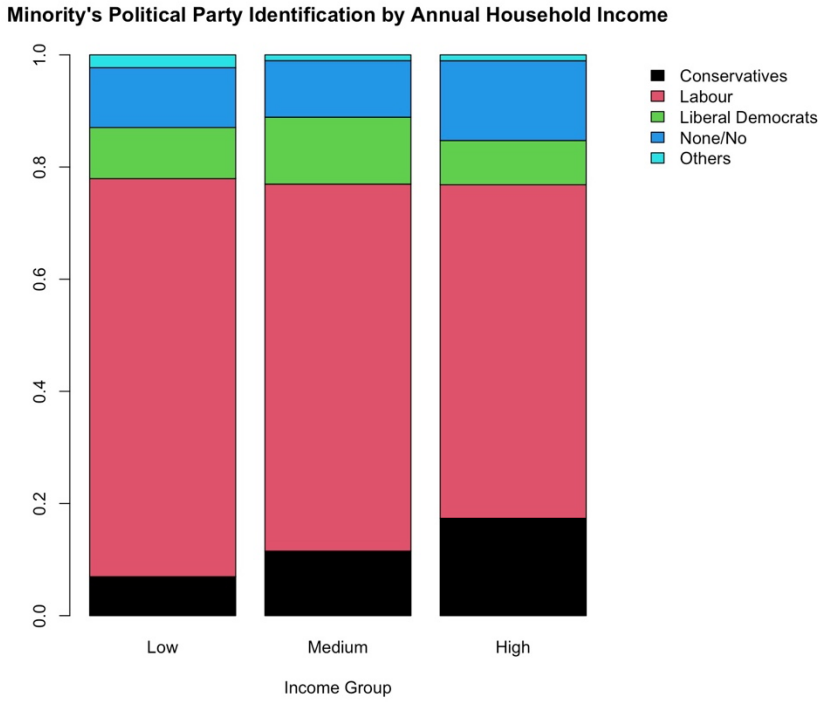

Figure 3: Minority's Political Party Identification by Income Group

Figure 3 presents a stacked bar chart showing the distribution of political party identification across income groups. The distribution appears relatively similar across the income categories, with the Labour Party consistently representing the largest proportion in each group, ranging from approximately 59% in the high-income group to about 71% in the low-income group, and only small variations observed for the other parties.

One noticeable pattern in the chart is that Conservative party identification increases as income increases, rising from about 7% in the low-income group to roughly 17% in the high-income group, while Labour party identification slightly decreases as income increases.

Overall, Figure 3 visually supports the Chi-square test results, indicating that although there is a statistically significant association between annual household income and political party identification among the five ethnic minority groups, the strength of the relationship is weak.

Discussion:

The results of the Chi-square test indicate that there is a statistically significant association between annual household income and political party identification among the five ethnic minority groups. However, the effect size measured by Cramér's V (0.103) suggests that the strength of this association is weak, indicating that income alone does not strongly determine political party preference within the sample.

Figure 3 illustrates the distribution of political party identification across income groups. Labour support remains dominant across all income levels, accounting for approximately 59% to 71% of respondents. While Labour identification decreases slightly as income increases, Conservative identification shows the opposite trend, rising from about 7% in the low-income group to around 17% in the high-income group. Despite these patterns, the overall distribution of party identification remains relatively similar across income groups, which is consistent with the weak association indicated by the statistical analysis.

Several limitations should be considered when interpreting these results. First, the dataset represents responses from a single year, which may not fully capture longer-term political trends. Second, the data are cross-sectional, meaning the analysis identifies associations but cannot establish a causal relationship between income and political party identification. Third, the analysis only considers income as a predictor, while other socio-demographic factors such as education, age, or regional differences may also influence political preferences. Finally, some categories were collapsed into broader groups to address sparse cells in the contingency table, which improved the validity of the Chi-square test but may have reduced some detail in the data.

Conclusion:

This study examined the relationship between annual household income and political party identification among five ethnic minority groups. The Chi-square test showed a statistically significant association between the two variables. However, the effect size indicated that the relationship is weak, suggesting that income alone plays a limited role in explaining political party identification within the sample. Overall, the findings indicate that while some income-related patterns exist, other factors are likely to have a greater influence on political preferences.

References:

1. Heath, A. et al. , British Election Study Ethnic Minority Survey, 2010 [computer file]. Colchester, Essex: UK Data Archive [distributor], April 2012. SN: 6970, http://dx.doi.org/10.5255/UKDA-SN-6970-1
2. Bobbitt, Z. (2021b, August 25). Chi-Square Test of Independence: Definition, formula, and example. Statology. https://www.statology.org/chi-square-test-of-independence/
3. Bobbitt, Z. (2021, August 14). The four assumptions of a Chi-Square test. Statology. https://www.statology.org/chi-square-test-assumptions/#:~:text=It's%20assumed%20that%20individuals%20can,expected%20value%20less%20than%201.
4. Grace-Martin, K. (2023, August 3). Observed Values less than 5 in a Chi Square test – No biggie. The Analysis Factor. https://www.theanalysisfactor.com/observed-values-less-than-5-in-a-chi-square-test-no-biggie/